

Towards fast and constant-time implementations of Hamming Quasi-Cyclic

A case study on ARM Cortex-M4 devices

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Overview

1. Introduction
2. Background
3. Implementation
4. Experimental results
5. Conclusions

Introduction

Classic computers *vs* classic cryptography

Symmetric cryptosystems

Key enumeration: perform an exhaustive search on the whole keyspace, $\mathcal{O}(n)$ time

Asymmetric cryptosystems

General number field sieve: factor a composite number n in $\mathcal{O}(e^{1.9(\log n)^{1/3}(\log\log n)^{2/3}})$ (subexponential) time

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Quantum computing: state of the art

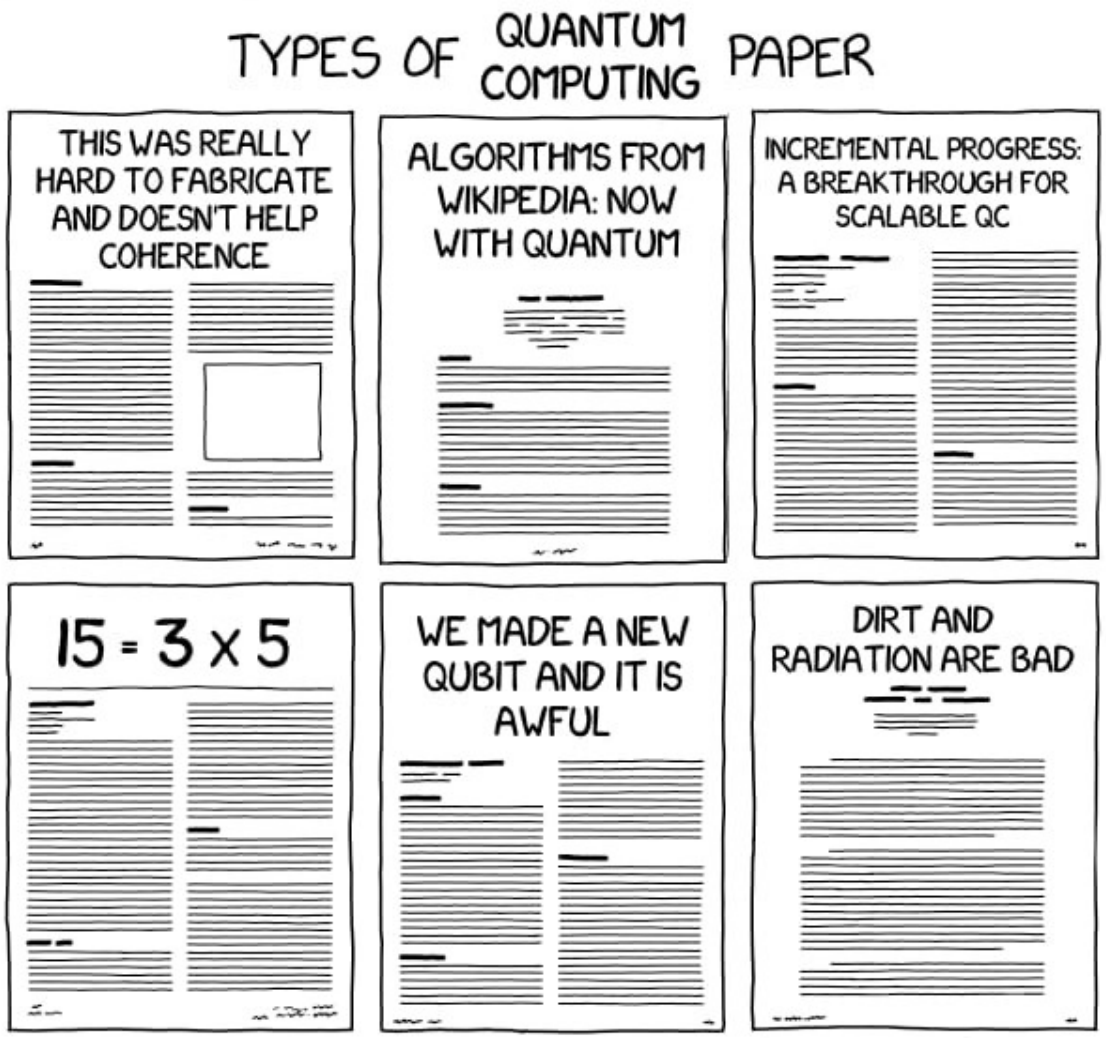


Figure: Types of Quantum Computing papers (credits: XKCD, Matt Reed)

Quantum computers *vs* classic cryptography

Symmetric cryptosystems

Grover's algorithm: search in a database of n elements in $\mathcal{O}(n^{1/2})$ time and $\mathcal{O}(\log n)$ space

Asymmetric cryptosystems

Shor's algorithm: factor a composite number n, on a quantum computer, in $\mathcal{O}((\log n)^2(\log\log n)(\log\log\log n))$ (polynomial) time [Sho94]

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NIST Post-Quantum Cryptography Standardization

Timeline

- 26/02/16: Announcement of NIST's call for submissions
- 21/12/17: Announcement of the first round candidates (69)
- 30/01/19: Announcement of the second round candidates (26)
- 22/07/20: Announcement of the third round candidates (7 finalists, 8 alternatives)
- 2022-2024: Publication of the standard drafts

Background

Implementation

Experimental results

References I



Peter W. Shor, *Algorithms for Quantum Computation: Discrete Logarithms and Factoring*, 35th Annual Symposium on Foundations of Computer Science, Santa Fe, New Mexico, USA, 20-22 November 1994, IEEE Computer Society, 1994, pp. 124–134.